Current Figure and Table Reference

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This file collects the current manuscript displays. Recovery is defined as

$$\rho = \frac{\mathrm{acc}(\mathrm{patch}) - \mathrm{acc}(\mathrm{receiver})}{\mathrm{acc}(\mathrm{donor}) - \mathrm{acc}(\mathrm{receiver})}. \tag{1}$$

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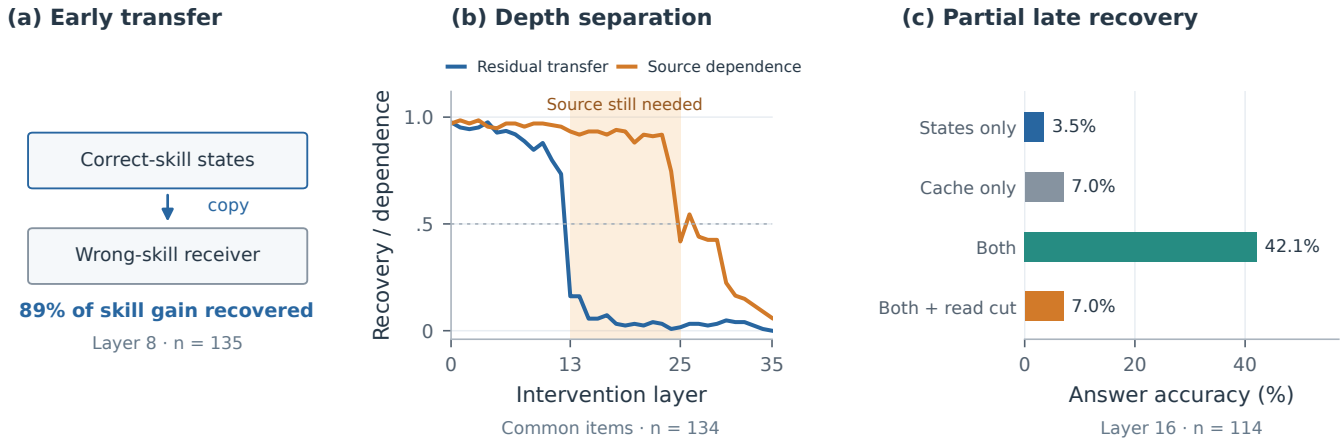


Figure 1: From effective early transfer to source-cache repair. (a) Early skill states recover most of the donor's benefit on the screened Qwen3-8B/MedCalc set. (b) On common eligible items, residual transfer weakens while source dependence remains high. Dependence is $1 - q(L)$, the fraction of rescued answers lost after cutting source access. (c) At layer 16, replacing source states together with their source cache recovers part of late transfer, and cutting subsequent reading removes most of that recovery. Panels use their indicated evaluation populations.

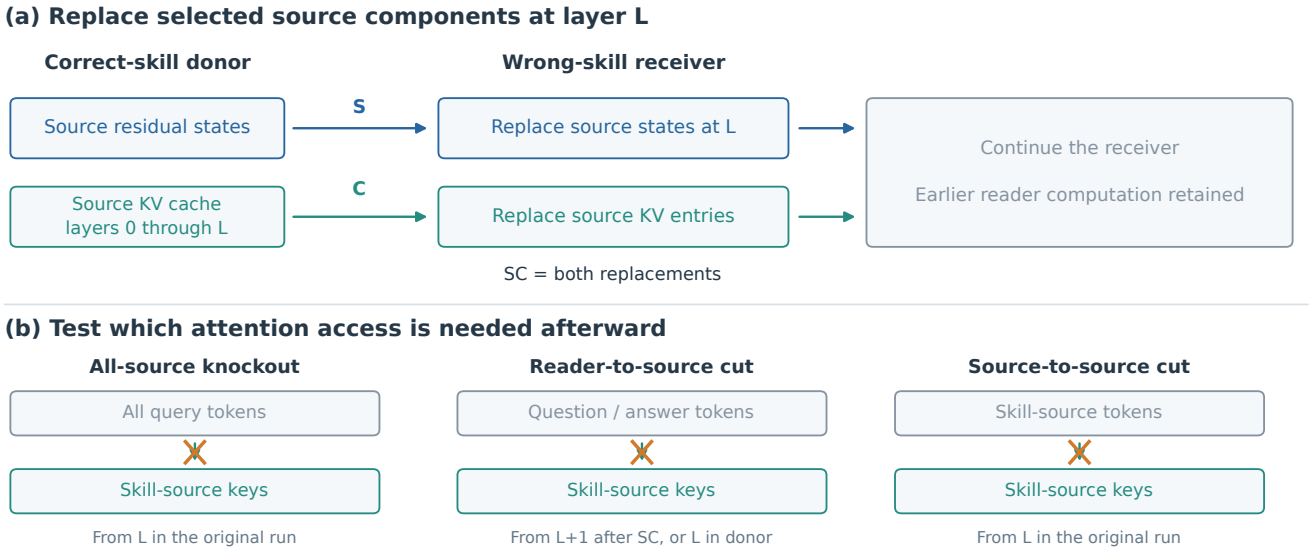


Figure 2: Intervention design. (a) The correct-skill donor supplies source residual states (S), source-position KV entries through layer L (C), or both (SC) to the wrong-skill receiver. Earlier reader computation is retained. (b) Attention cuts distinguish access by reader tokens from updates within the skill source. After SC, the reader cut begins at $L + 1$. In the original correct-skill run, path cuts begin at L. All cuts continue through the remaining prompt processing and decoding.

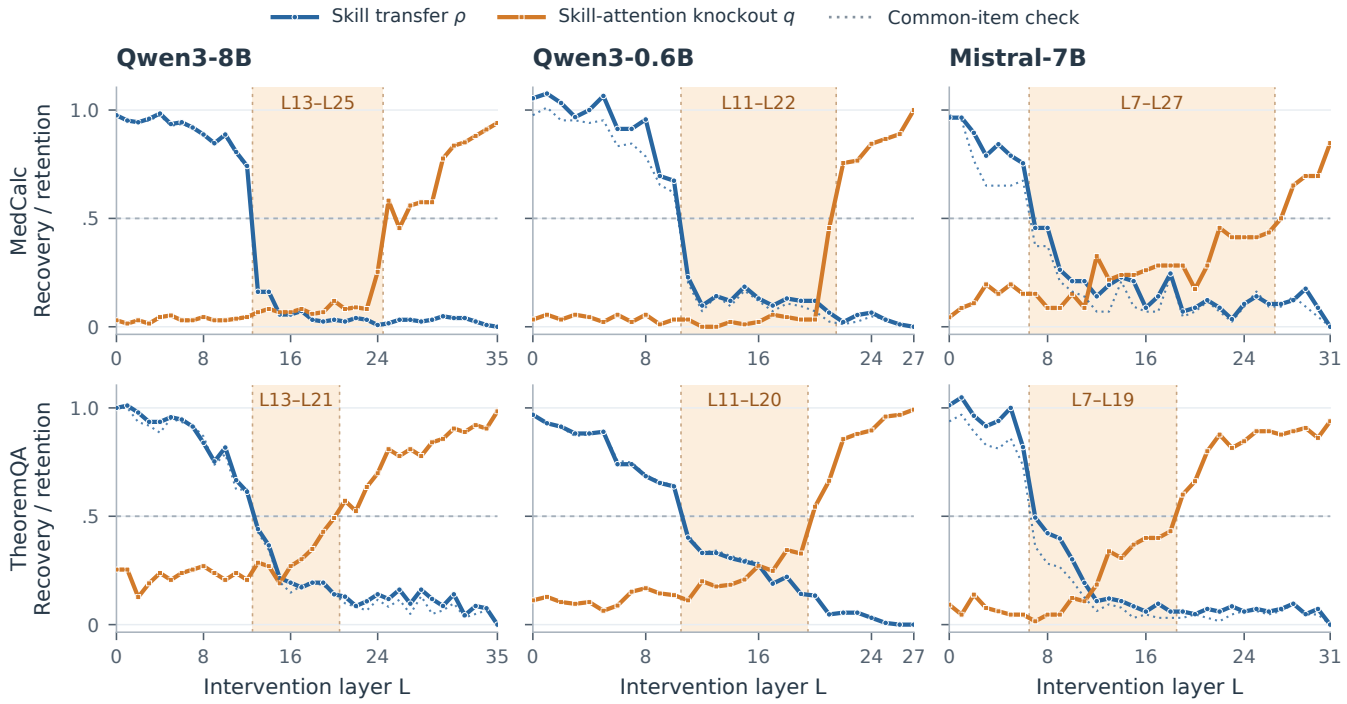


Figure 3: Depth profiles across three models and two tasks. Blue shows skill-span transfer recovery $\rho(L)$ in a wrong-skill receiver. Orange shows the retained fraction $q(L)$ after cumulative attention knockout in the correct-skill execution. Dotted traces repeat the tests on common eligible questions. Shading spans the first half-effect boundaries, with their layers marked above each panel. Every plotted layer is measured.

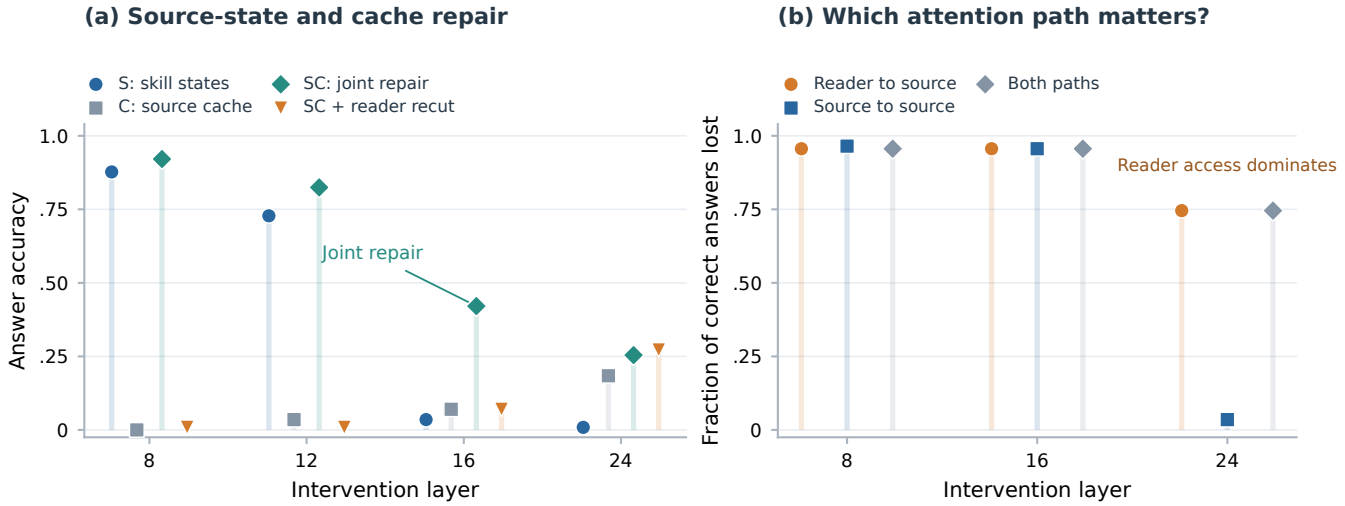


Figure 4: Source-state/cache replacement and attention-path tests on the same 114 Qwen3-8B/MedCalc items from 14 skills. (a) Accuracy following residual replacement (S), cache replacement (C), joint repair (SC), or joint repair followed by a reader cut from the next layer. (b) Fraction of donor-correct answers lost when reader-to-source, source-to-source, or both attention paths are cut from the indicated layer onward. Points are measured estimates, horizontally offset within each layer for legibility. Table 3 reports uncertainty for the main contrasts.

Table 1: Early transfer controls, Qwen3-8B/MedCalc, layer 8: 135 items from 14 calculators. Accuracy and recovery ρ are point estimates. Final-position patches replace fewer states than the skill-span patches.

Intervention	Accuracy	ρ
Correct skill donor	.919	1.00
Wrong skill receiver	.000	.00
Correct skill span	.815	.89
Last prompt position	.030	.03
Last 16 prompt positions	.059	.06
Other skill content	.044	.05
Correct span, permuted	.044	.05
Norm-matched noise	.022	.02
Identity patch	.000	.00

Table 2: Boundary layers on common eligible items. T/R **gives the first half-effect crossings for transfer and attention retention, respectively.** $\Delta = R - T$ **measures their separation in layers. The last column gives its skill-group bootstrap** 95% **confidence interval.**

Task	Model	n	T/R	Δ	95% CI
MedCalc	Qwen3-8B	134	13/25	12	[12.0, 17.0]
MedCalc	Qwen3-0.6B	90	11/22	11	[10.0, 13.0]
MedCalc	Mistral-7B	46	7/27	20	[10.0, 24.0]
TheoremQA	Qwen3-8B	63	13/21	8	[5.0, 10.0]
TheoremQA	Qwen3-0.6B	125	11/20	9	[8.0, 10.0]
TheoremQA	Mistral-7B	65	7/19	12	[9.0, 13.0]

Table 3: Focused mechanism contrasts, 114 items from 14 skills. Differences are in percentage points (pp), with skill-group bootstrap 95% confidence intervals. The additive interaction is $SC - S - C +$ receiver.

Contrast	Difference	95% CI
L16: $SC - S$	+38.6	[26.6, 46.5]
L16: additive interaction	+31.6	[14.1, 41.6]
L16: $SC - (SC +$ recut$)$	+35.1	[23.2, 43.0]
L24: reader minus self loss	+71.1	[53.3, 86.5]

Table 4: Full-dataset rerun against the development subset, Qwen3-8B on MedCalc, layer 8. Recovery is Eq. 1; entries show point estimates. *per-calc* is the main text's restriction, *per-item* the weaker one. Controls reach 0.12 under the group screen and 0.25 under the item screen; each arm uses its own matched baselines.

arm	full, per-calc (n = 135)	subset (n = 60)	full, per-item (n = 369)
transplant `real`	**0.89**	0.87	0.94
same skill, matched prefix	0.65	0.73	0.84
half dose $\alpha = 0.5$	0.04	0.06	0.11
positions permuted	0.05	0.08	0.21
norm-matched noise	0.02	0.10	0.18
other calculator	0.05	—	0.25
other family, mean	0.05	0.06	0.16
same family, mean	0.12	0.05	0.24
identity patch	0.00	0.00	0.00
rank 128 / 64 / 32 / 16	0.77/0.63/0.12/0.05	0.77/0.54/0.13/0.06	0.88/0.70/0.22/0.18
bottom-64 / bottom-16	0.03/0.04	0.00/0.06	0.11/0.12
quarters $q_0/q_1/q_2/q_3$	0.09/**0.56**/0.12/0.05	0.12/0.56/0.12/0.12	0.18/0.64/0.22/0.19